



Optimizing Food Delivery Operations: A Data-Driven Analysis of Delivery Delays

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zomato

Founded in 2008

Industry: Food Delivery

Presence: Multiple Indian Cities

Zomato is one of India's largest online food delivery and restaurant discovery platforms. It connects customers, restaurants, and delivery partners through a technology-driven marketplace, enabling users to discover restaurants, place food orders, and receive deliveries at their doorstep.

Revenue Sources

Restaurant comission

Delivery fees

Advertisements

Membership

Business Success Depends on Fast & Reliable Deliveries
Delivery time directly impacts customer satisfaction, repeat orders, and operational efficiency.

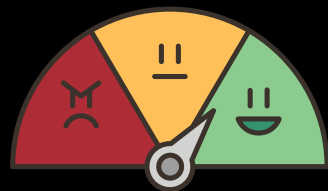
Problem statement

Fast delivery time is critical for Zomato as it directly drives customer retention, ensures food arrives at optimal eating temperatures, and maintains a competitive edge over rival services.

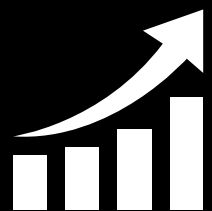
“How can Zomato reduce delivery delays by identifying the key operational factors affecting delivery time?”



Longer Delivery Time



Lower Customer Satisfaction



Reduced Operational Efficiency

Approach



Data



Feature engineering



Exploratory Data Analysis



Statistical Testing



Business Recommendations

Dataset and Feature Engineering

Source: Kaggle Zomato Delivery Operations Dataset

Objective: Analyze factors influencing food delivery time.

[Link to dataset](#)

20 Columns

45000+ rows

Target: delivery time

- **ID:** Unique identifier for each delivery.
- **Delivery_person_ID:** Unique identifier for delivery person.
- **Delivery_person_Age:** Age of the delivery person.
- **Delivery_person_Ratings:** Ratings given to delivery person.
- **Restaurant_latitude:** Latitude of the restaurant.
- **Restaurant_longitude:** Longitude of the restaurant.
- **Delivery_location_latitude:** Latitude of the delivery location.
- **Delivery_location_longitude:** Longitude of delivery location.
- **Order_Date:** Date of the order.
- **Time_Ordered:** Time the order was placed.

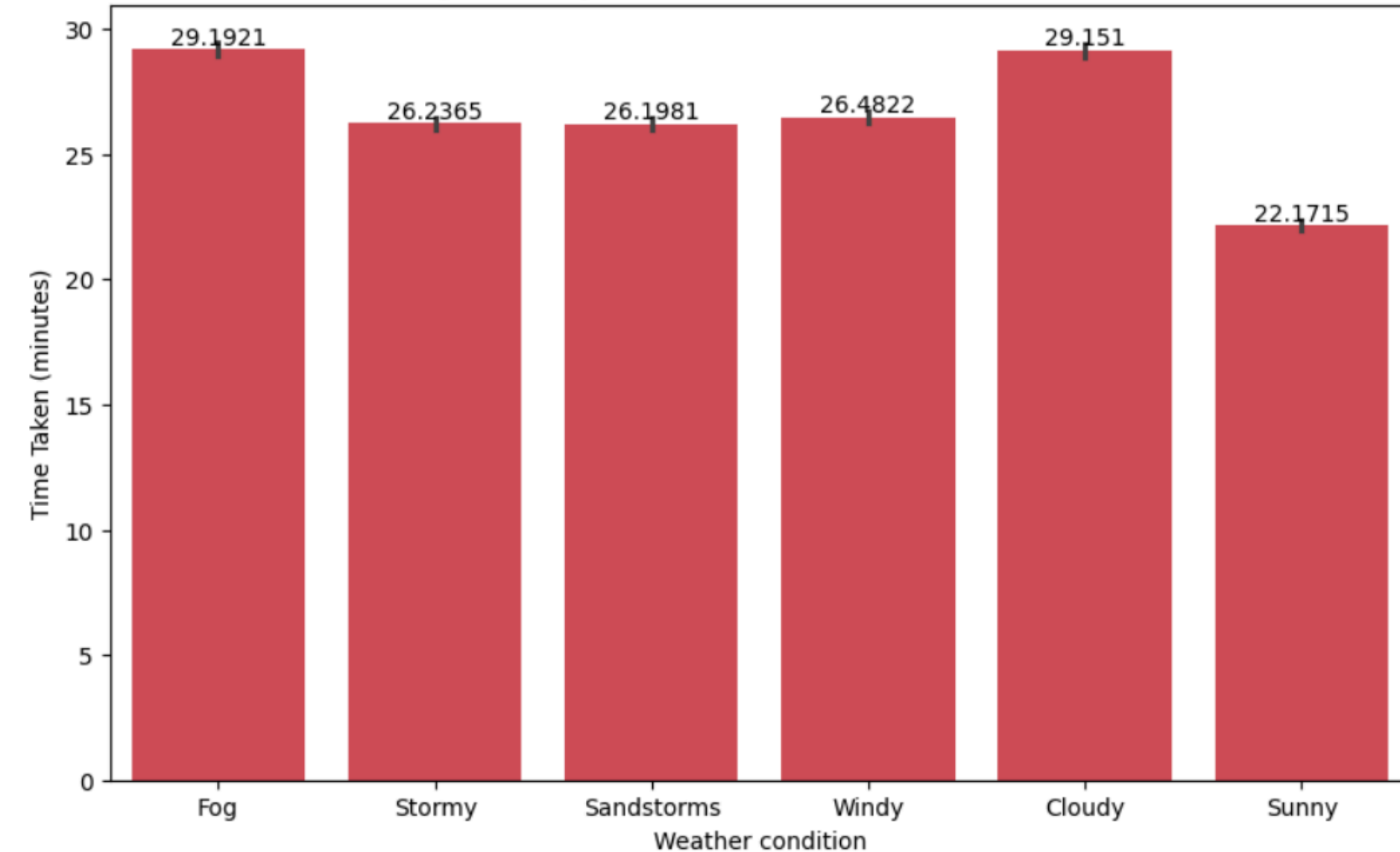
- **Time_Order_picked:** Time the order was picked up for delivery.
- **Weather_conditions:** Weather conditions at the time of delivery.
- **Road_traffic_density:** Density of road traffic during delivery.
- **Vehicle_condition:** Condition of the delivery vehicle.
- **Type_of_order:** Type of order (e.g., dine-in, takeaway, delivery).
- **Type_of_vehicle:** Type of vehicle used for delivery.
- **Multiple_deliveries:** Indicator of no. of deliveries.
- **Festival:** Indicator of whether the delivery was during a festival.
- **City:** City where the delivery took place.
- **Time_taken (min):** Time taken for delivery in minutes.

New Features Created

- **order_month:** Month in which order was placed.
- **deliver_person_age_bin:** Age group of delivery person
- **peak_hours:** Indicator of the peak time of the day.
- **pickup_delay_min:** Difference of time between placing and picking

Exploratory Data Analysis

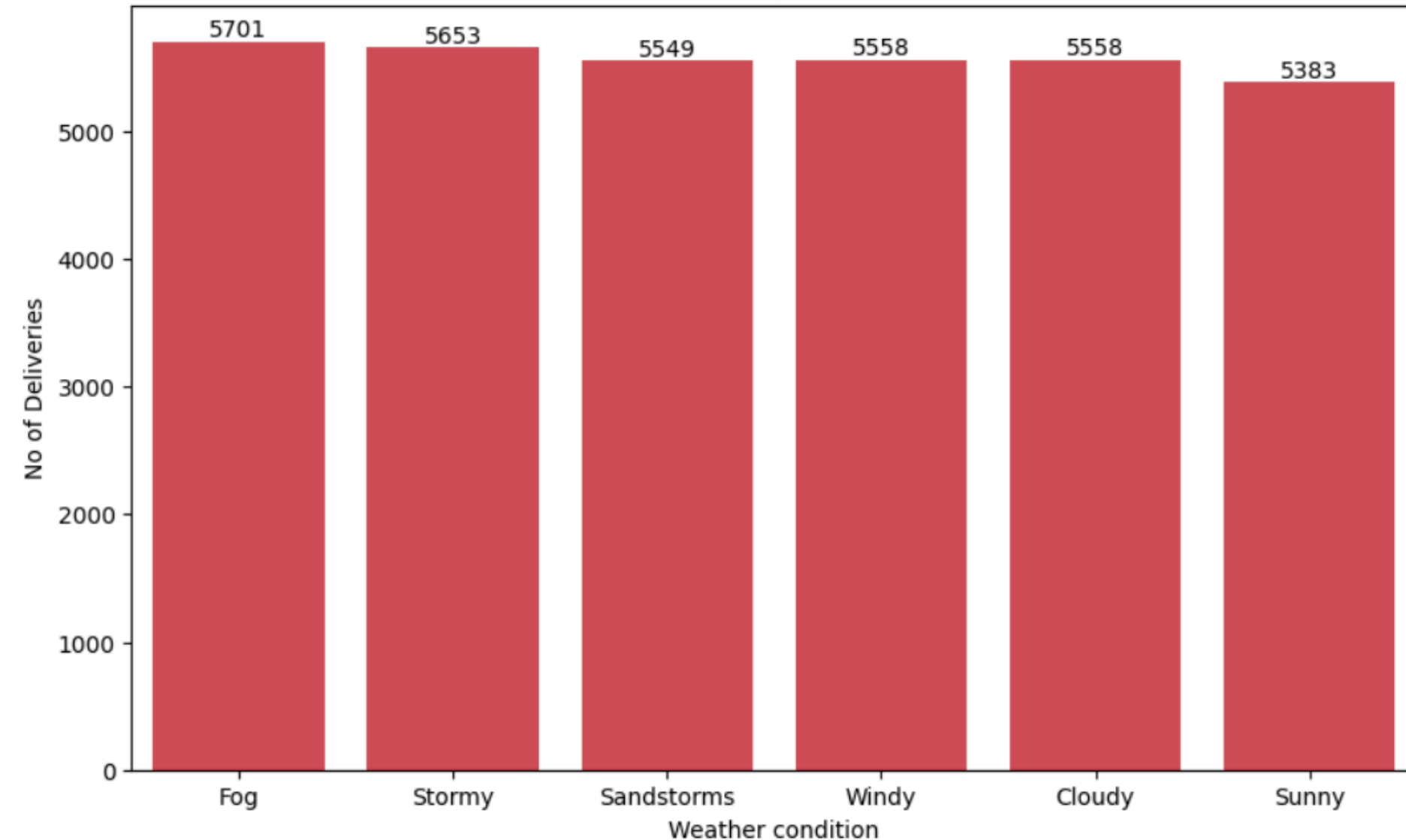
Average Delivery Times across different Weather Condition



Inference: Average delivery time during foggy weather condition was the highest followed by cloudy. The minimum average time was for sunny day.

Comment: As it is difficult to deliver food on Fog and cloudy weather conditions, hence the delivery time is longer whereas sunny day provides no problem to delivery partners.

Count of Deliveries by Weather Conditions

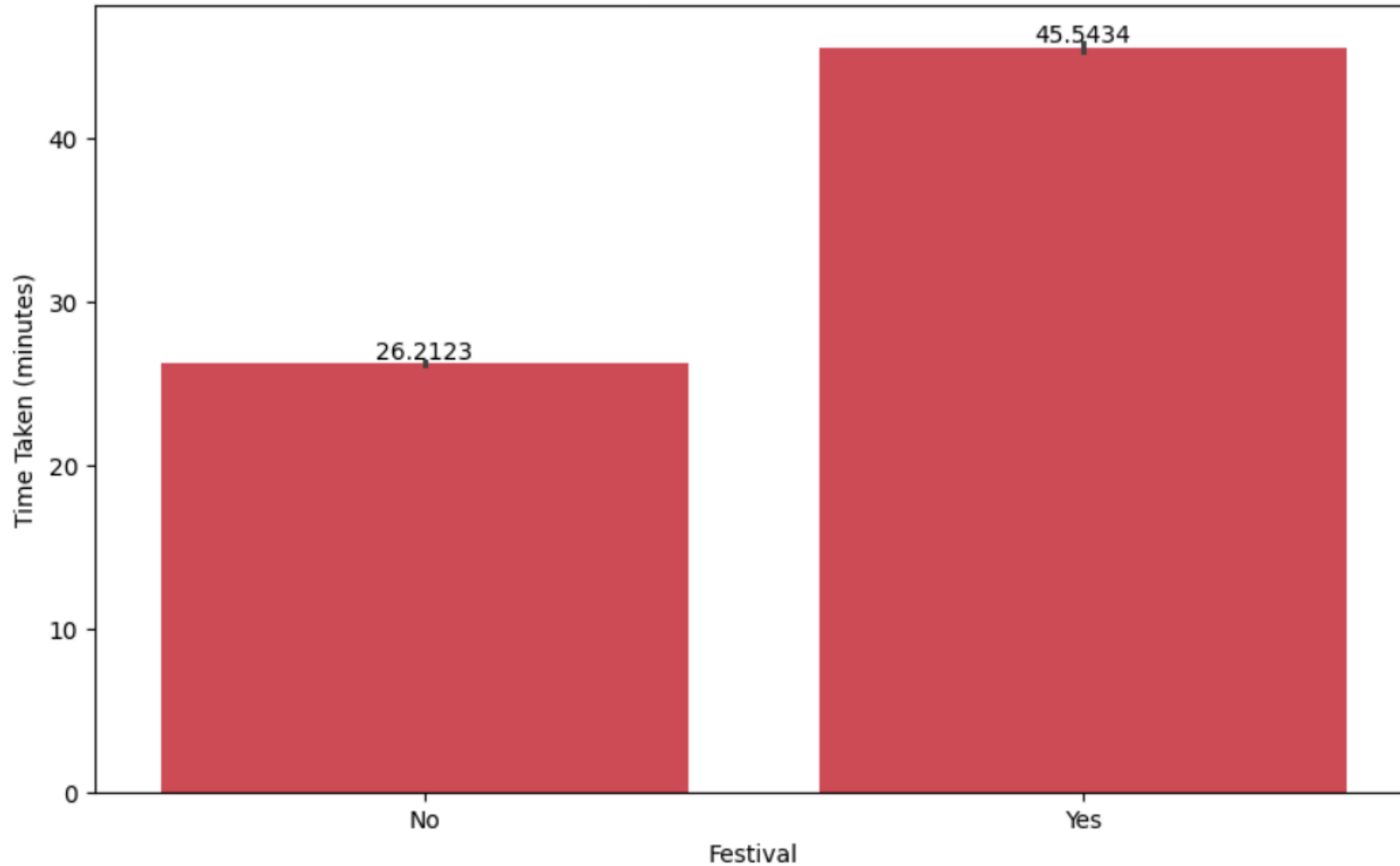


Inference: Highest number of orders were placed during Foggy weather condition whereas least number of orders were placed during sunny day.

Comment: Customers tend to stay in the home during Fog, stormy and sandstorm conditions, while more people go out to eat at the restaurant on a sunny clear day.

Exploratory Data Analysis

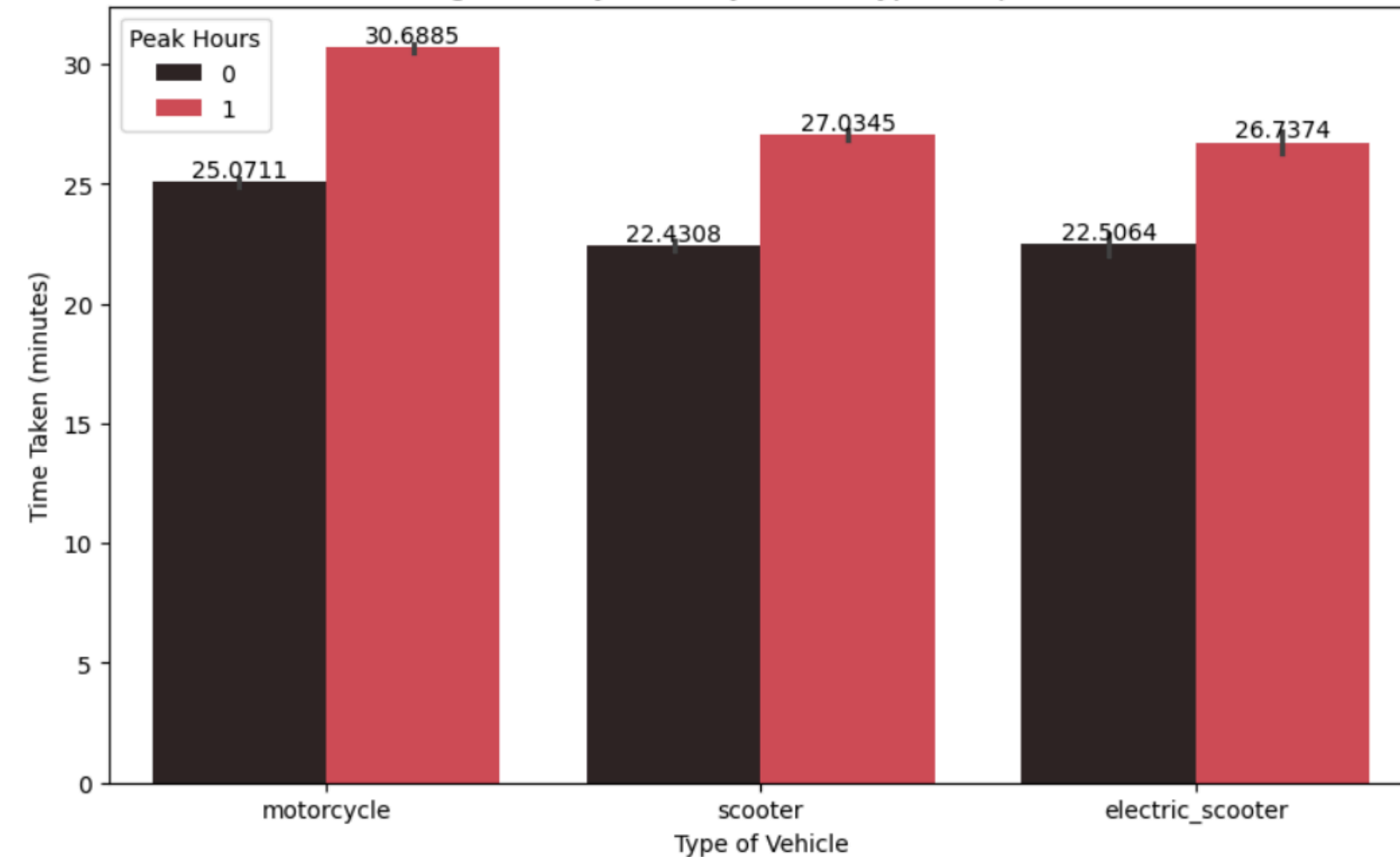
Average Delivery Times depending upon Festival



Inference: Average delivery time during festivals is around 45 mins whereas the time for delivery during normal days is just around 26 mins.

Comment: As there is high demand in restaurants and for online delivery during festivals, the delivery time is very high compared to a normal day.

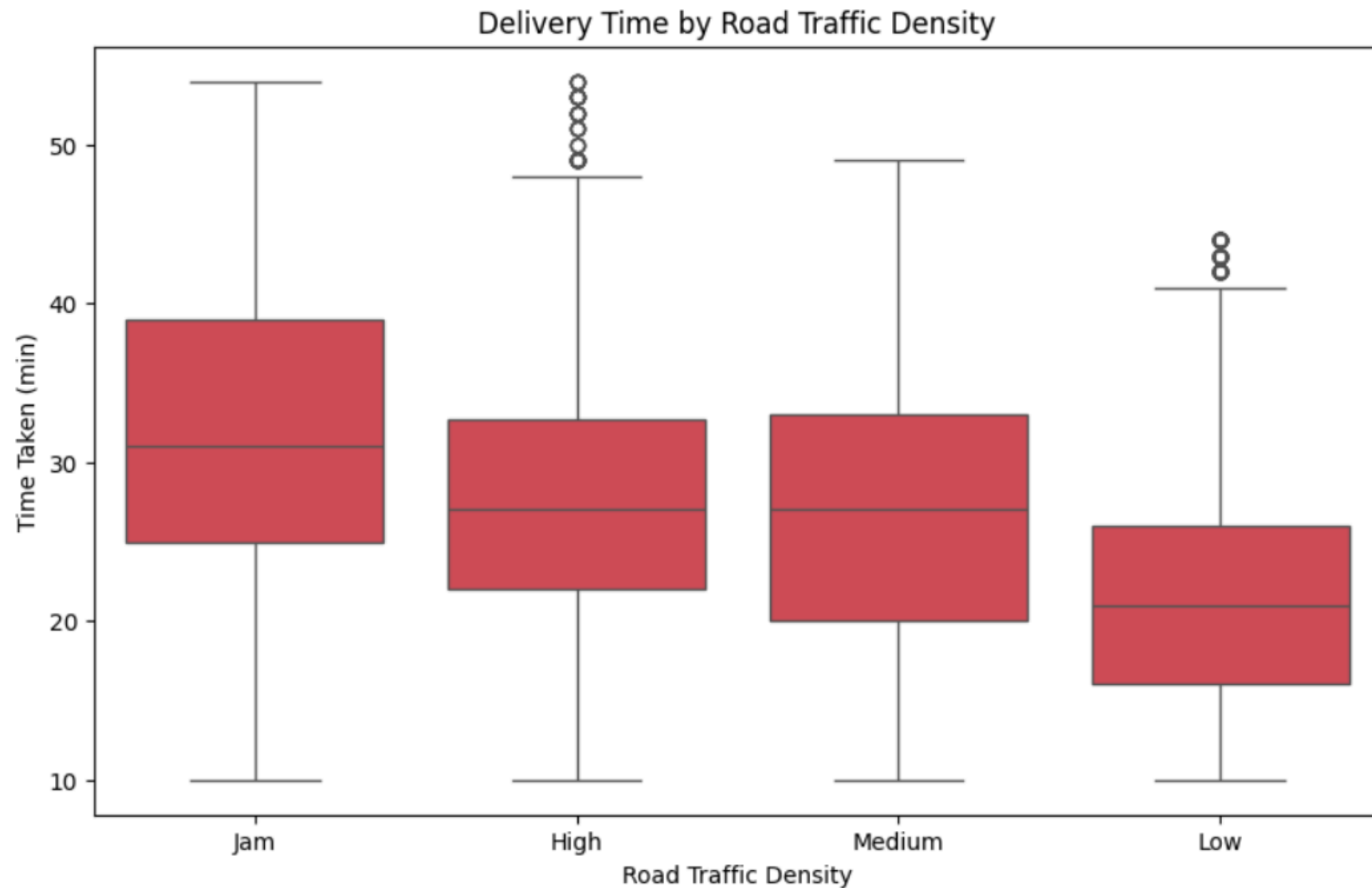
Average Delivery Times by Vehicle Type and peak hours



Inference: Average delivery time for the delivery partner with motorcycle is the highest whereas for the ones with electric scooter, it is lowest.

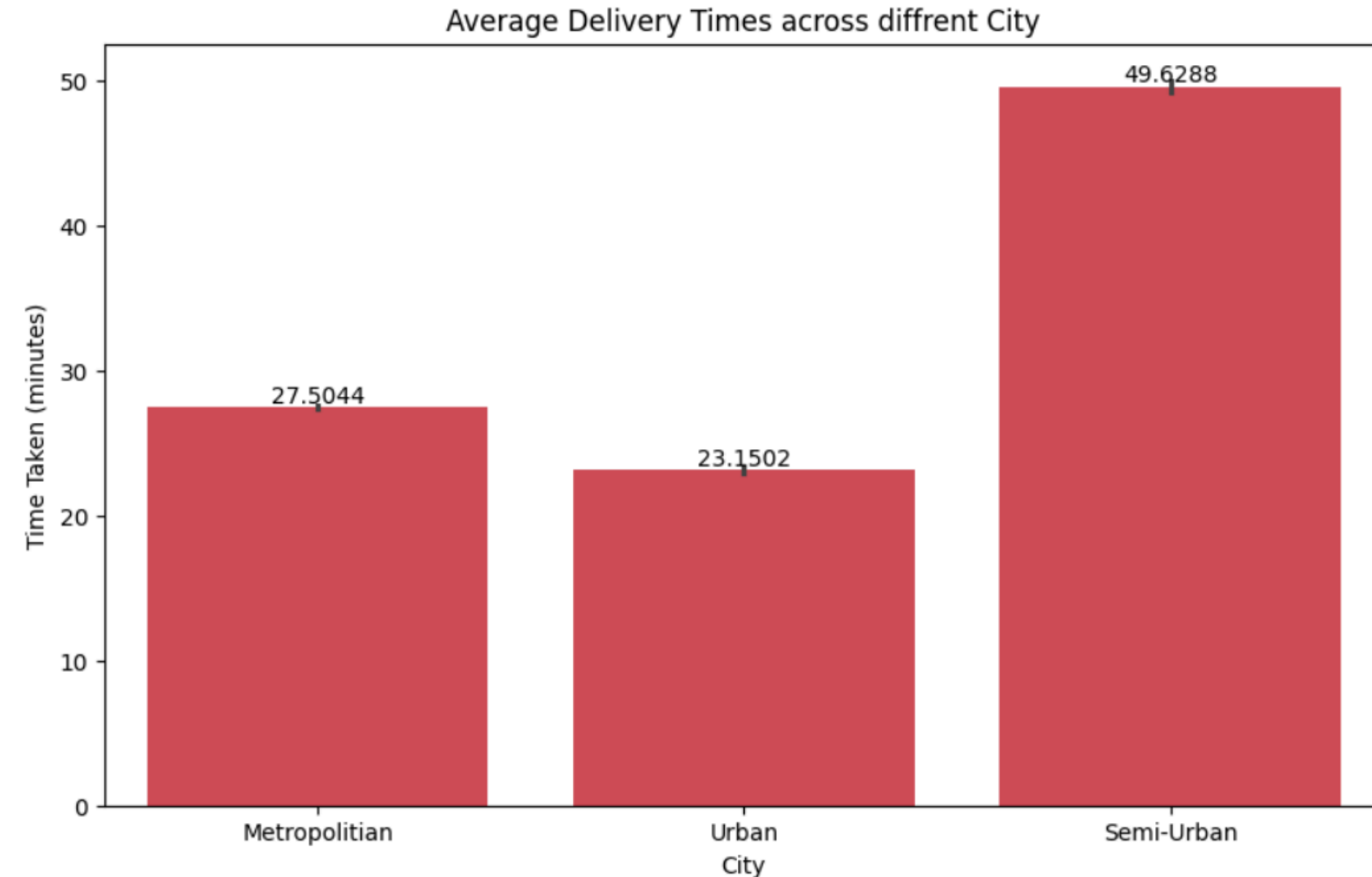
Comment: Electric scooters are highly agile, lightweight, and require zero gear shifting, allowing riders to swiftly weave through congested traffic.

Exploratory Data Analysis



Inference: The median time for Jam traffic condition is highest whereas it is lowest for low traffic conditions.

Comment: Majority of the time is wasted in traffic during jam conditions hence avg. delivery time is highest whereas low traffic provides good opportunity for delivery partner to deliver food on time.



Inference: Average delivery time in semi-urban cities is considerable high as compared to other cities.

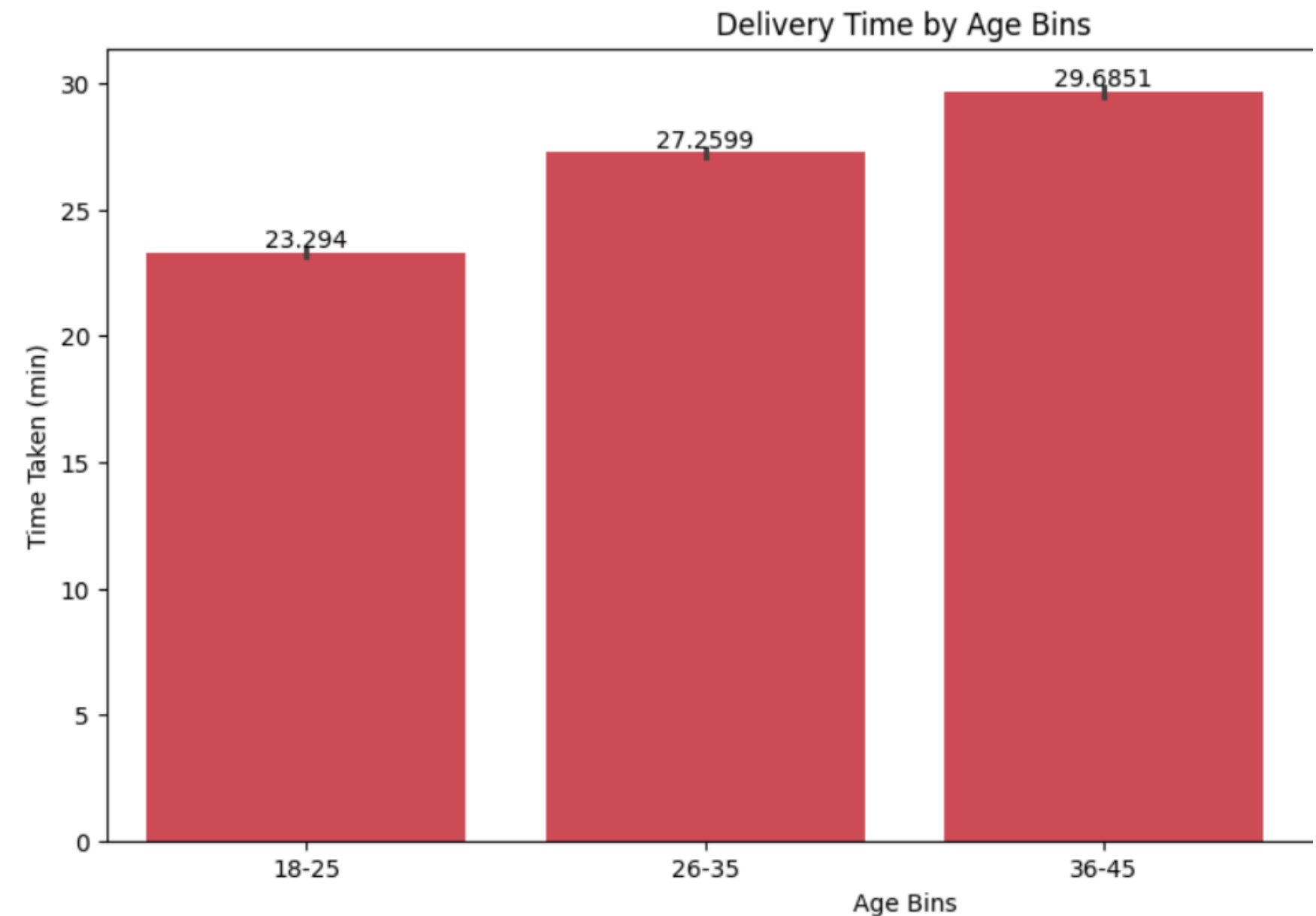
Comment: Semi-urban cities in India generally experience the highest average Zomato delivery times due to lower restaurant density, larger delivery radii per order, and less advanced kitchen infrastructure.

Exploratory Data Analysis



Inference: Delivery partners with higher rating have delivered much before as compared to the ones who have low rating.

Comment: Customers provide good feedback to delivery partners who deliver on time or early and vice versa.



Inference: Average delivery time for delivery partners is increasing with increasing age groups.

Comment: Young partners have higher risk tolerance, less fatigue and they are very comfortable with using the maps and other app features.

Statistical testing

Hypothesis 1: Delivery times are shorter for higher-rated delivery persons.

H0: There is no difference in average delivery times between high-rated and low-rated delivery persons.

H1: Higher-rated delivery persons have shorter average delivery times.

Approach:

- Split the data into two groups: high-rated (above the median) and low-rated (below or equal to the median) delivery persons.
- Perform a t-test to compare the average delivery times.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- There is a significant difference in average delivery times between high-rated and low-rated delivery persons. Specifically, higher-rated delivery persons tend to have shorter average delivery times. This suggests that higher ratings may correlate with better performance, possibly due to factors such as experience, efficiency, or customer service skills.

Statistical testing

Hypothesis 2: Delivery times differ by weather conditions.

H0: There is no difference in average delivery times across different weather conditions.

H1: Average delivery times differ based on weather conditions.

Approach:

- This is a comparison between a categorical variable and a numeric variable, with the categorical variable having more than two categories.
- Use ANOVA to compare delivery times across different weather categories.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- There is a significant difference in average delivery times based on weather conditions. This suggests that various weather conditions impact delivery efficiency, indicating that some conditions may lead to longer delivery times than others.

Statistical testing

Hypothesis 3: Older delivery persons have longer delivery times.

H0: There is no difference in average delivery times between young (younger than 30) and old (30 or older) delivery persons.

H1: There is a significant difference in average delivery times between young and old delivery persons.

Approach:

- This is a comparison between a categorical variable (age groups) and a numeric variable (delivery time). Use a two-sample t-test.
- Create age groups (e.g., younger than 30 and 30 or older) and perform a t-test to compare average delivery times between these groups.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- There is a significant difference in average delivery times between young and old delivery persons. This suggests that age may impact delivery performance.

Statistical testing

Hypothesis 4: Delivery times are significantly affected by road traffic density.

H0: There is no difference in average delivery times across different levels of road traffic density.

H1: There is a significant difference in average delivery times across different levels of road traffic density.

Approach:

- This is a comparison between a categorical variable (road traffic density levels) and a numeric variable (delivery time).
- Use ANOVA to assess differences in average delivery times among various traffic density categories.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- There is a significant difference in average delivery times across different levels of road traffic density. This indicates that traffic conditions play a crucial role in determining delivery efficiency, with varying densities affecting how quickly deliveries are completed.

Statistical testing

Hypothesis 5: Delivery times are longer during festivals.

H0: There is no difference in avg. delivery times for orders during festivals and non festival days.

H1: Avg. delivery times are longer for deliveries made during festivals compared non-festival periods.

Approach:

- This is a comparison between a categorical variable (festival indicator) and a numeric variable (delivery time).
- Use a t-test to assess differences in average delivery times between deliveries made during festivals and those made during non-festival periods.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- This suggests that delivery times are indeed significantly longer during festival periods, potentially due to increased traffic, higher demand, or other contributing factors.

Statistical testing

Hypothesis 6: Different types of vehicles have different average delivery times.

H0: There is no difference in average delivery times among different types of vehicles.

H1: There is a significant difference in average delivery times among different types of vehicles.

Approach:

- This is a comparison between a categorical variable (type of vehicle) and a numeric variable (delivery time).
- Use ANOVA (Analysis of Variance) to assess differences in average delivery times among various types of vehicles.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- Average delivery times vary significantly by vehicle type, suggesting that some vehicles are more efficient for delivery.

Statistical testing

Hypothesis 7: Delivery times vary significantly by city.

H0: There is no difference in average delivery times among different cities.

H1: There is a significant difference in average delivery times among different cities.

Approach:

- Similar to the vehicle analysis, this hypothesis compares a categorical variable (city) with a numeric variable (delivery time).
- Again, ANOVA is an appropriate method to use here.

Results:

- $p_value < 0.05$
- Reject Null Hypothesis
- The analysis reveals that average delivery times vary significantly across different cities, suggesting that factors such as traffic conditions, infrastructure, and local regulations may affect delivery efficiency.

Business Recommendations

Workforce Optimization (Based on Hypothesis 1 and 3)

“Higher-rated and more experienced delivery partners complete deliveries faster”

Performance-Based Order Allocation

Allocate more high-priority orders to top-rated riders during peak hours.

Impact:

- Faster deliveries
- Better customer experience

Rider Training Program

Analyze behaviors of top performers and create training modules for lower-rated riders.

Impact:

- Increase fleet efficiency
- Reduce delivery variability

Rating-Based Incentives

Reward consistently high-performing riders to incentivize them to perform better.

Impact:

- Improve rider retention
- Better service quality

Business Recommendations

Demand & Traffic Management (Based on Hypothesis 2, 4 and 5)

“External conditions significantly impact delivery performance.”

Dynamic Fleet Positioning

Move riders closer to high-demand zones before rush hours.

Weather-Aware ETA

Adjust delivery ETAs dynamically based on weather conditions.

Festival Surge Planning

Deploy additional riders during festivals. Introduce temporary incentive programs.

Impact:

- Reduced delays during peak demand
- Better ETA accuracy
- Improved customer trust

Business Recommendations

Operational Efficiency (Based on Hypothesis 6 and 7)

“Delivery performance varies across vehicle types and cities.”

Vehicle Optimization

Assign vehicle types based on distance and locality.

City-Specific Operations

Develop city-level delivery strategies instead of one universal model.

Hyperlocal Routing

Optimize rider assignment using city-specific traffic patterns.

Impact:

- Improved operational scalability
- Reduced delivery time
- Better fleet utilization

THANK YOU

“Your favorite food, right on time, every time.”